# Chern number

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# Overview


1. Derivation of Hall conductance
   - Linear response theory
   - DC conductance
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2. Berry phase, Berry connection, Berry curvature
3. Real space Chern number
4. Tests on Haldane's honeycomb model

# Linear response theory

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- Kubo's formula:

$$\delta\langle A\rangle_t = \langle A\rangle_t - \langle A\rangle_0 = -i\sum_n{}' \int_{t_0}^{t} dt' \langle n|[A_I(t), V_I(t')]|n\rangle$$

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- Set $t - t_0 = \infty$,

$$\sigma_{\alpha\beta}(\omega) = \frac{i}{\omega}\sum_{n\neq m}{}' \frac{\langle n|J_\alpha|m\rangle\langle m|J_\beta|n\rangle}{\omega + E_n - E_m} - \frac{\langle n|J_\beta|m\rangle\langle m|J_\alpha|n\rangle}{w + E_m - E_n}$$

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- Note that:

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- DC conductance:

$$\sigma_{\alpha\beta} = \lim_{\omega\to 0} Re\ \sigma_{\alpha\beta}(\omega) = i\hbar\sum_{n\neq m}{}' \frac{\langle n|J_\beta|m\rangle\langle m|J_\alpha|n\rangle - (\alpha\leftrightarrow\beta)}{(E_n-E_m)^2}$$

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- Finally we get the Hall conductance:

$$\sigma_{xy} = \frac{e^2}{\hbar} C_1$$

where $C_1$ is called the first Chern number of band insulators, with

$$C_1 = \int \frac{d^2k}{(2\pi)^2}\mathcal{F}, \quad \mathcal{F} = \sum_n{}' i\Big[\langle \partial_{k_x} n|\partial_{k_y} n\rangle - (x \leftrightarrow y)\Big]$$

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- In 1982, Thouless, Kohmoto, Nightingale and den Nijs showed that $C_1$ is an integer in gapped crystal with periodic boundary condition, which explain IQHE, hence $C_1$ is also called TKNN integer.

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$$\dot{\theta}_n(t) = -\frac{1}{\hbar}E_n(t) + i\langle n(t)|\partial_t|n(t)\rangle,$$

$$\theta_n(t) = -\frac{1}{\hbar}\int_0^t dt' E_n(t') + \gamma_n,$$

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- Only for a closed loop in $R$ space, it becomes a gauge invariant, called Berry phase, and Berry connection is also defined

$$\gamma_n = \oint_C A_n(R)\cdot dR, \quad A_n(R) = i\langle n(R)|\nabla_R n(R)\rangle$$

# Berry phase, Berry connection, Berry curvature

- Berry curvature is defined as the same form in EM theory:

$$A_\mu^{(n)} = i\langle n|\partial_\mu n\rangle, \quad F_{\mu\nu}^{(n)} = \partial_\mu A_\nu^{(n)} - \partial_\nu A_\mu^{(n)} = i\Big[\langle \partial_\mu n|\partial_\nu n\rangle - (\mu \leftrightarrow \nu)\Big]$$

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- Mathematically, for a fibre bundle with base space $M$, gauge group $G$, the total Chern class and nth Chern class are given by:

$$c(F) := \det(I + \frac{iF}{2\pi}) = \sum_{n=0} c_n(F), \quad c_n(F) = 0 \text{ when } 2n > \dim M \tag{1}$$

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- Chern character and Chern number are defined as:

$$\mathrm{ch}_n(F) = \frac{1}{n!}\,\mathrm{Tr}\,(\frac{iF}{2\pi})^n, \quad C_n = \int_M \mathrm{ch}_n(F)$$

# Alternate form

- Define projection to the states below the Fermi level: $P = \sum_n' |nk\rangle\langle nk|$, the Berry curvature and Chern number can be expressed as:

$$\mathcal{F} = PdP \wedge dP,$$

$$C_1 = \frac{1}{2\pi i}\int_{\mathbb{T}^2} \mathrm{Tr}\,\mathcal{F} = \frac{1}{2\pi i}\int_{\mathbb{T}^2} dk_1 dk_2\, \mathrm{Tr}\, P[\partial_1 P, \partial_2 P]$$

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- An $2n$ dimensional generation is:

$$C_n = \frac{1}{n!(2\pi i)^n}\int_{\mathbb{T}^n} (\prod_{i=1}^{2n} dk_i) \sum_\sigma (-1)^\sigma\, \mathrm{Tr}(P \prod_{i=1}^{2n} \partial_{\sigma_i} P)$$

# Real space Chern number

- For coordinate and momentum, $\partial_i P = [X_i, P]$, so nth real space Chern number is:

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- In 2 dimension,

$$C_1 = 2\pi i \operatorname{Tr} P\big[[X_1, P], [X_2, P]\big]$$

# Real space Chern number

- First shown by Kitaev: $\nu = 2\pi i \operatorname{Tr} P\big[[X_1, P], [X_2, P]\big]$, he also showed that [Kitaev, 2005]:

$$\det(U_1 U_2 U_1^\dagger U_2^\dagger) = \exp(-(2\pi)^2 \operatorname{Tr}[PX_1P, PX_2P]),\ U_j = e^{2\pi i P X_j P}$$

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- Which is equivalent to the Bott index proposed by Loring and Hastings [Loring, Hasting, 2010]:

$$C_b = \frac{1}{2\pi} \operatorname{Im} \operatorname{Tr} \ln(V_1 V_2 V_1^\dagger V_2^\dagger), \quad V_j = P e^{2\pi i X_j} P$$

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- $L \to \infty, \nu = C_b \to C_1$; [Ge, Rigol, 2018]

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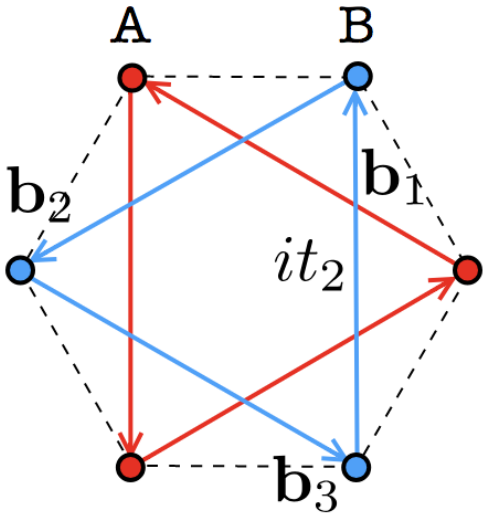
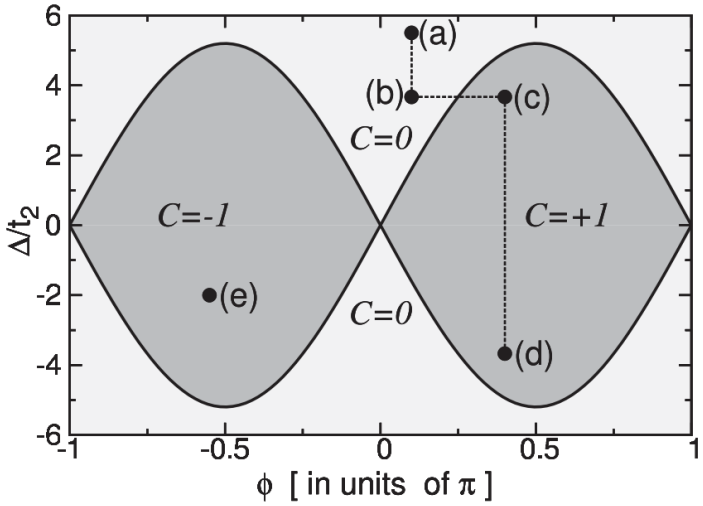
- $L \to \infty, \nu = C_b \to C_1$; [Ge, Rigol, 2018]
- For finite $L$, $\nu_{edge} + \nu_{bulk} = 0$; [Bianco, Resta, 2011] (anomaly cancellation)

# Test on Haldane's model

- Following Bianco's method, local Chern number $\nu = \sum_r \nu(r)$
- Haldane's model:

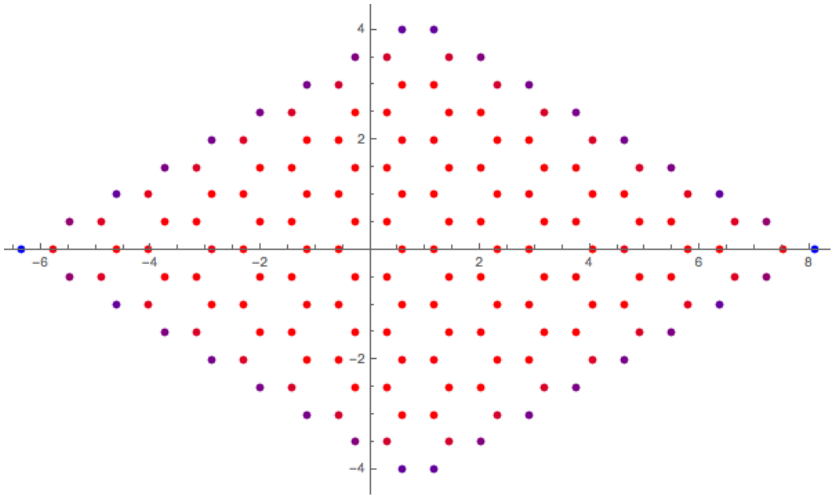
$$H = -t_1 \sum_{\langle i,j \rangle} c_i^\dagger c_j + t_2 e^{i\phi} \sum_{\langle\langle i,j \rangle\rangle} c_i^\dagger c_j + \Delta \sum_{i \in A} c_i^\dagger c_i - \Delta \sum_{i \in B} c_i^\dagger c_i$$

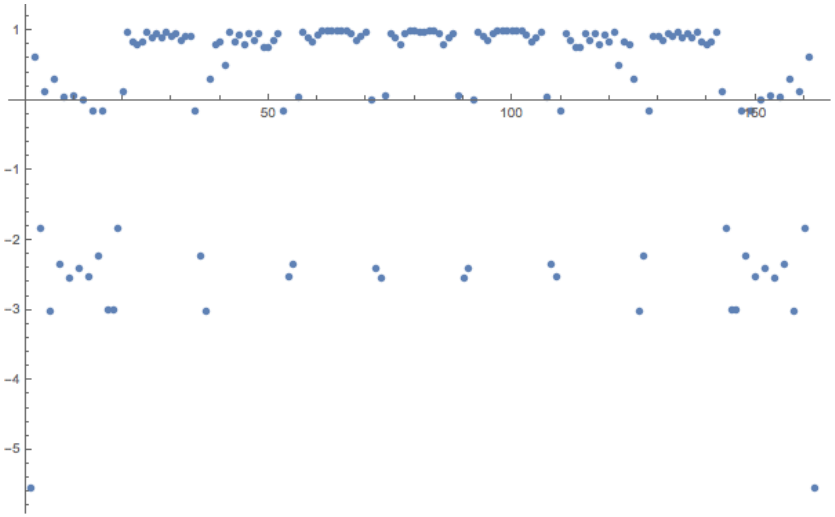


# Test on Haldane's model

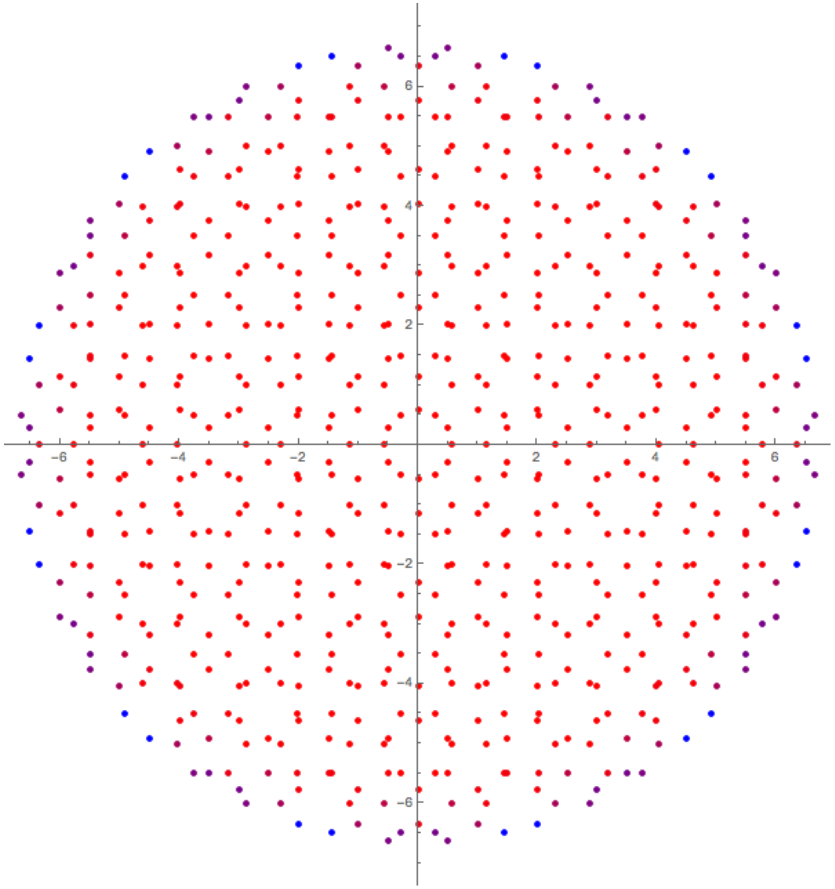
- $t_1 = 0.1, t_2 = 0.33, \Delta = 0, \phi = \pi/2, 7 \times 8$, cell average is made

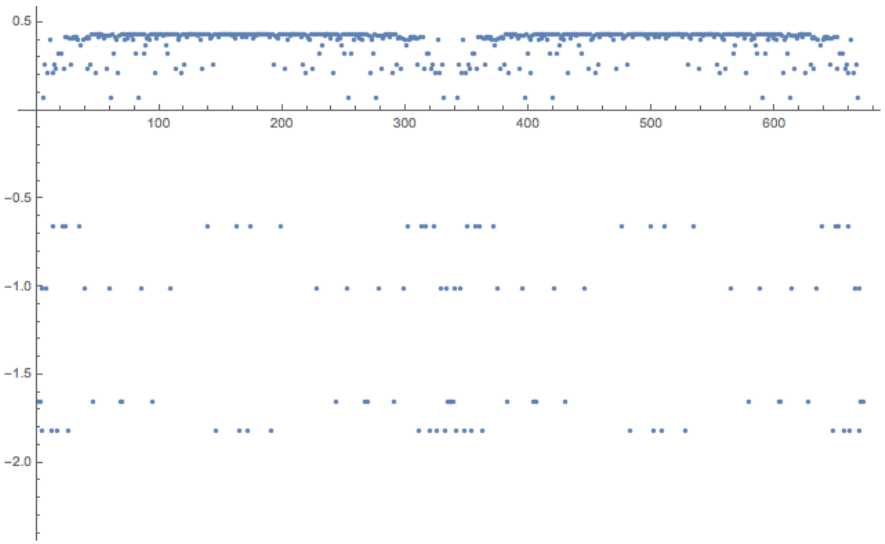


# Test on bilayer Haldane's model

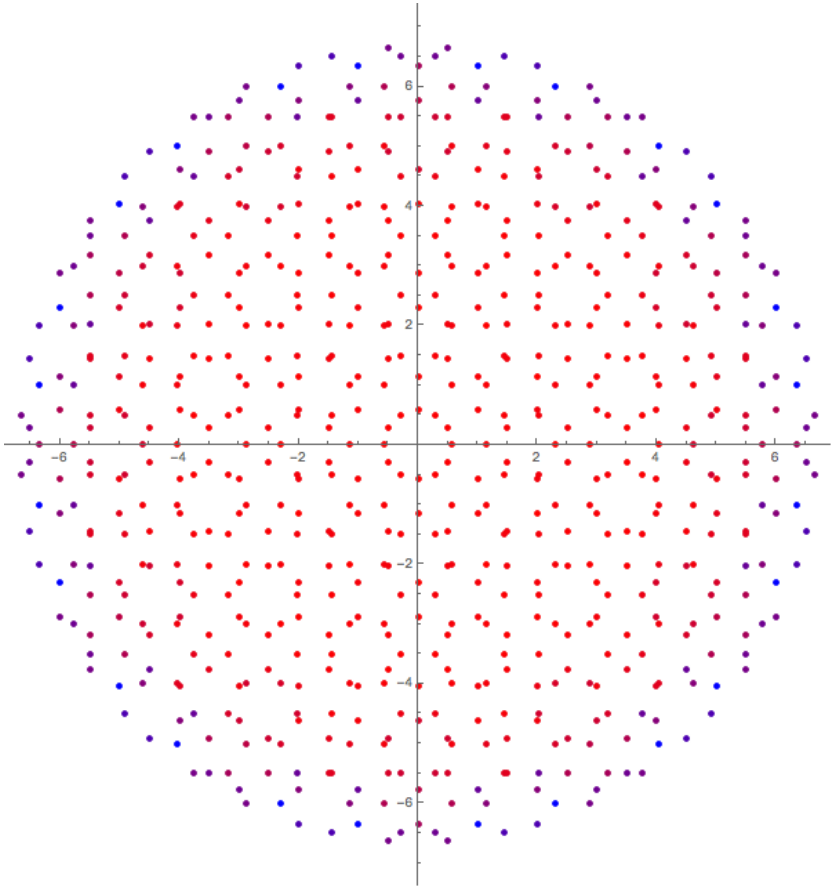
- $t_3 = 0.1$

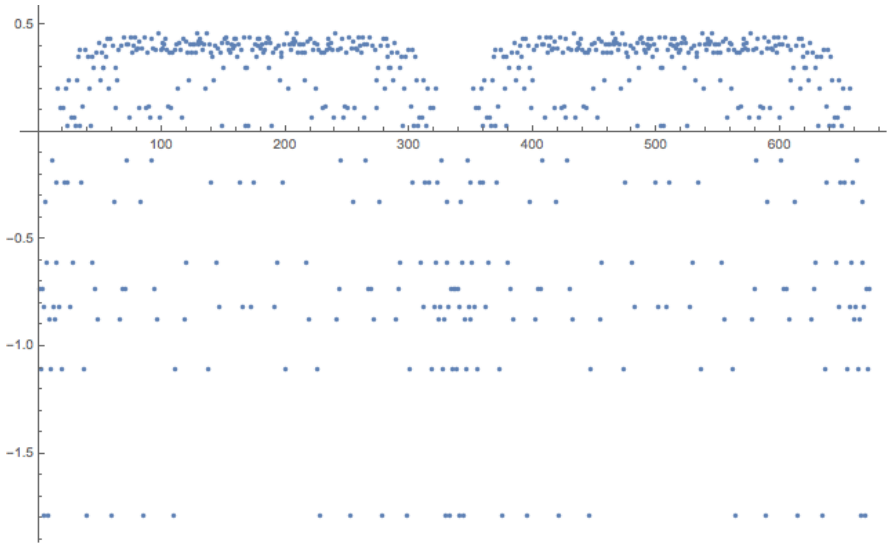


# Test on bilayer Haldane's model

- $t_3 = 1$



# Summary

## Chern number in reciprocal space

$$C_1 = \int \frac{d^2k}{(2\pi)^2}\mathcal{F}, \quad \mathcal{F} = \sum_n{}' i\Big[\langle \partial_{k_x} n | \partial_{k_y} n\rangle - (x \leftrightarrow y)\Big]$$

## Chern number in real space

$$\nu = 2\pi i \,\mathrm{Tr}\, P\big[[X_1, P], [X_2, P]\big]$$

$$C_b = \frac{1}{2\pi}\mathrm{Im}\,\mathrm{Tr}\ln(V_1 V_2 V_1^\dagger V_2^\dagger), \quad V_j = P e^{2\pi i X_j} P$$

What's not included in this review:

- Spin Chern number $\rightarrow$ quantum spin Hall;
- Application to topological insulators;
- Nontrivial band topology $\rightarrow$ edge states...

# References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- Yang Ge and Marcos Rigol, *Topological phase transitions in finite-size periodically driven translationally invariant systems*, Phys. Rev. A 96, 023610 (2017)
- Raffaello Bianco and Raffaele Resta, *Mapping topological order in coordinate space*, Phys. Rev. B 84, 241106 (2011)

Suggest reading:

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